

DEGLUTITION / SWALLOWING

Headings

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- **Definition** : Refers to the passage of food from the oral cavity to the stomach.
- **Comprises 3 Stages** :—
 - a. **Oral stage** : Mouth → Pharynx
 - b. **Pharyngeal stage** : Pharynx → Esophagus
 - c. **Esophageal stage** : Esophagus → Stomach

1. ORAL STAGE

- **Nature** : Voluntary phase.
- **Mechanism** :—

Bolus of food formed after mastication



Placed over dorsum of tongue



Tongue forces bolus into the oropharynx by pushing up and back against hard palate

2. PHARYNGEAL STAGE

- **Nature** : Involuntary phase caused by the swallowing reflex.
- **Duration** : Entire process takes about 1 — 2 seconds.

Stimulus: Food bolus enters pharynx

Receptors: Swallowing receptor areas of pharynx (*especially on tonsillar pillars*)

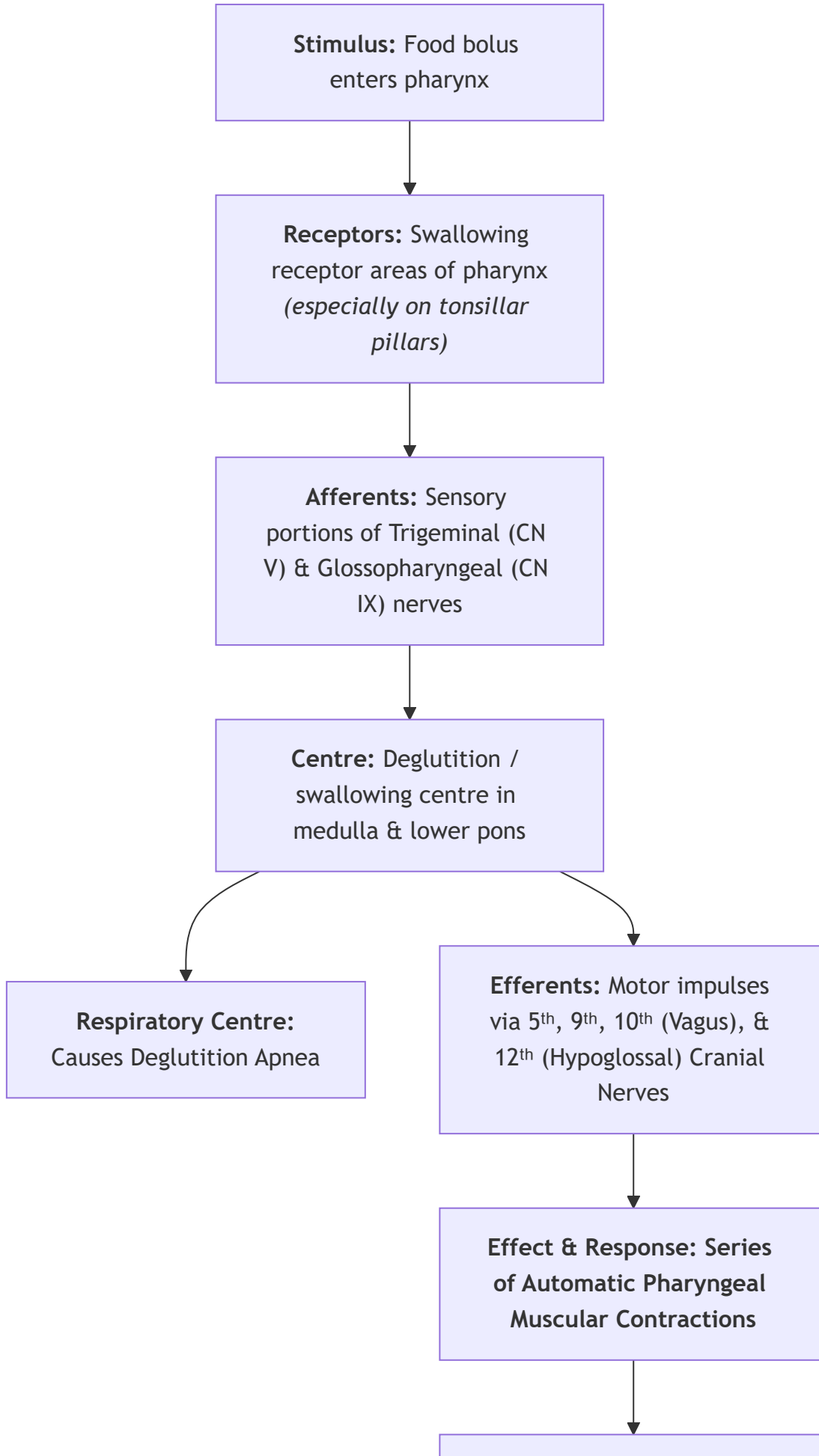
Afferents: Sensory portions of Trigeminal (CN V) & Glossopharyngeal (CN IX) nerves

Centre: Deglutition / swallowing centre in medulla & lower pons

Respiratory Centre:
Causes Deglutition Apnea

Efferents: Motor impulses via 5th, 9th, 10th (Vagus), & 12th (Hypoglossal) Cranial Nerves

Effect & Response: Series of Automatic Pharyngeal Muscular Contractions



1. Soft palate closes
posterior nares / nasal
cavities
2. Vocal cords come closer
→ Larynx closed
3. Larynx pulled upward &
forward
4. Pharyngoesophageal
sphincter relaxes
5. Food passes to
esophagus via Rapid
Peristaltic Wave

3. ESOPHAGEAL STAGE (3RD STAGE)

- **Nature** : Involuntary stage.
- **Function** : Esophagus forms a passage for movement of the food bolus from the pharynx to the stomach.
- **Movement Type** : Peristaltic waves (*wave of contraction followed by wave of relaxation of muscle fibers traveling in an aboral direction*).
- **2 Types of Peristaltic Contractions** :—
 - a. Primary peristaltic contractions
 - b. Secondary peristaltic contractions

A. Primary Peristaltic Contractions

- When the bolus reaches the upper part of the esophagus, **primary (1^o) peristalsis starts**.
- Peristaltic contractions pass down through the rest of the esophagus, propelling the bolus towards the stomach.
- **Intraluminal Pressure Changes** :—
 - **Initially** : Pressure becomes negative (*due to stretching of closed esophagus by elevation of larynx*).
 - **Immediately after** : Pressure becomes positive ($\uparrow$ *increases up to 10 – 15 cm of H₂O*) as the contractile wave sweeps through.

B. Secondary Peristaltic Contractions

- If primary contractions are **unable to propel the bolus** into the stomach (*e.g., bolus gets lodged*):

Secondary peristaltic contractions appear



Induced by local distension of upper esophagus by lodged bolus



Contractions pass down, producing positive (+ve) pressure to clear the lumen

ROLE OF LOWER ESOPHAGEAL SPHINCTER (LES)

